Core Java Study Plan Day 8

Programming Questions

1. Write a Java program to demonstrate Encapsulation by creating a Student class with private data members name and age, and public getter and setter methods.

2. Write a Java program to demonstrate Inheritance using the extends keyword. Create a Parent class with a method display() and a Child class that inherits and calls the method.

3. Write a Java program to demonstrate Method Overriding and Runtime Polymorphism using a parent class Animal and child classes Dog and Cat.

4. Write a Java program to demonstrate the use of an Interface. Create an interface Vehicle with a method start() and implement it in a Car class.

5. Write a Java program using an Abstract class. Create an abstract class Shape with an abstract method area(), and implement it in a Circle class.

6. Write a Java program to demonstrate Method Overriding where a child class provides its own implementation of a method defined in the parent class.

7. Write a Java program to demonstrate Multiple Inheritance using Interfaces. Create two interfaces Printable and Showable and implement both in a single class.

8. Write a Java program to demonstrate Data Hiding using the private access modifier. Create a BankAccount class with a private balance and methods to deposit and display the balance.

9. Write a Java program to demonstrate Abstraction using an abstract class Employee with an abstract method calculateSalary(), and implement it in a Developer class.

10. Write a Java program to demonstrate the use of the super keyword. Create a parent class with a variable and method, and access both from the child class using super.

Puzzles

1. Four people need to cross a bridge at night. Their crossing times are 1, 2, 7, and 10 minutes. They have one torch, at most two can cross at once, and the torch must be carried. What is the minimum total time?

2. You have a 5-litre container and a 3-litre container, with no markings, plus unlimited water. How can you measure exactly 4 litres?

3. A room has four walls and one window on each wall. A person standing inside can see only three windows. How is this possible?

4. A man lives on the 20th floor. Every morning he takes the lift to the ground floor. When returning, he goes only to the 12th floor and walks up, except on rainy days when he goes directly to the 20th. Why?

5. There are 100 lockers, all initially closed. In round 1 every locker is toggled, in round 2 every second locker, in round 3 every third locker, and so on up to round 100. Which lockers remain open?

Aptitude Questions

1. From 8 employees, select a team of 3 and designate one selected member as leader. How many ways?

A) 56 B) 112 C) 168 D) 336

2. How many 4-digit numbers can be formed using 1,2,3,4,5,6 without repetition that are divisible by 5?

A) 48 B) 54 C) 60 D) 72

3. Two fair dice are thrown. What is the probability that the sum is prime?

A) 5/12 B) 1/3 C) 7/18 D) 1/2

4. A bag has 5 red, 4 blue and 3 green balls. Two are drawn without replacement. Probability both are blue?

A) 1/11 B) 1/9 C) 2/11 D) 1/6

5. A fair coin is tossed 4 times. Probability of at least 3 heads?

A) 3/16 B) 5/16 C) 1/4 D) 7/16

6. Tea costing ₹240/kg is mixed with tea costing ₹300/kg in ratio 2:3. Find mixture cost per kg.

A) ₹264 B) ₹270 C) ₹276 D) ₹280

7. A 40-L mixture contains milk and water in ratio 7:3. How much water should be added to make ratio 7:5?

A) 8 L B) 10 L C) 12 L D) 14 L

8. A vessel contains 60 L acid-water mixture in ratio 2:1. How much mixture should be removed and replaced with water so final ratio is 1:1?

A) 10 L B) 15 L C) 20 L D) 25 L

9. Pointing to a woman, Raj said: 'Her mother is the only daughter of my mother's mother.' How is the woman related to Raj?

A) Sister B) Daughter C) Niece D) Aunt

10. A is brother of B. C is mother of A. D is father of C. E is daughter of D. How is E related to B?

A) Sister B) Aunt C) Cousin D) Mother